Hedging the Singularity

Abstract

AI stocks trade at extraordinary valuations. We develop an asset pricing model in which investors use AI stocks to hedge against an AI singularity that displaces their consumption. Because markets are incomplete—investors cannot trade private AI capital—AI stocks command a premium. Market incompleteness distorts both valuations and the efficient development of AI, creating a rationale for government transfers that becomes compelling when singularity-driven growth overwhelms deadweight costs. This paper was generated by AI, using `https://github.com/redacted/ralph-wiggum-asset-pricing/`.

Preface

This preface is mostly human-written. Please forgive typos and errors.

- Andrew Chen, April 2026.

Background

In March 2025, I thought I should write a paper about hedging the singularity.

I was worked up. I was using AI to prove theorems, do lit reviews, and vibe code in my daily life. Six months ago, I had thought each of these things is impossible.

What will happen in the next six years?! Will my entire job be replaced by AI? I had no idea.

But I did know that if there are huge AI disruptions, then tech stocks would most likely benefit. So if anything bad happens to my human capital, I could partially hedge.

I asked a friend if he would be interested in working on this paper. Unfortunately, he was busy. So, I thought I should use AI to do it.

This project was inspired by Novy-Marx and Velikov (2026) and Chris Lu et al. (2024). These projects use AI to generate massive amounts of academic research. My goal differs in quality over quantity. I want to generate just one paper, but one paper that (I hope) people find is worth reading.

This project was also inspired by Garleanu, Kogan, and Panageas's (2012) beautiful model of innovation and displacement risk. I read Garleanu et al. back when I was a PhD student and it has stuck with me.

Last, I drew from Hadfield-Menell and Hadfield (2018) and Bengio (2023), who apply ideas from economics to AI catastrophe risk. Hadfield-Menell and Hadfield (2018) explains the connection between incomplete contracting and AI alignment. Bengio (2023) frames AI catastrophe risk in terms of what I would call decision theory and human incentives---though the essay is written in plain English.

Previously, I dismissed "AI safety" as a politicized, activist cause. I still don't like the term "AI safety." "Safety" is such an excuse for exercising power over others.

But then the nature of AI changed. Things were progressing much faster than I expected. The Jan 15, 2025 episode of Machine Learning Street Talk with Yoshua Bengio left an impression on me. Bengio talked about AI catastrophe risk with no activism, no fear mongering. It was a straight, rational discussion of the seriousness of AI catastrophe risk.

The Paper Generation Process (April 2026)

The paper is generated by the ralph-wiggum-asset-pricing Github repo. My goal was "human as Clockmaker": the human designs the paper generation algorithm, and then the AIs write the paper, unattended. This would be a clear demonstration of AI displacement risk.

To this end, I tried to generate the paper with (Geoff Huntley's) "Ralph Wiggum Loop":

```
Clockmaker Ralph Loop:
    1. An author-plan agent generates an improvement plan based on previous
       test results, a paper specification, and other supporting docs
       (e.g. key literature)
    2. An author-improve agent executes `improvement-plan.md`
    3. Test agents evaluate the improved paper and code
    4. If any test fails, go back to 1.
```

Like Ralph Wiggum (from the Simpsons), the agents don't seem to mind doing the same thing, over and over again. Unlike Ralph, the agents won't keep going unless you feed them money.

With unlimited money, this Ralph loop is all you need. If the output is bad, just add more tests.

But after adding 25 tests, I would still find random minor problems (e.g. Opus used SIC codes to characterize AI stocks, why?!). At this point, the Ralph loop was eating straight through two $200/month Claude Code subscriptions.

So I ultimately went with a hybrid approach. I would first "set the direction" of the loop, by running the author agents five times, independently. Then I'd pick the best draft, and initialize the loop with it. Even then, I had to edit two sentences for accuracy after the run ended.

The first 5 or so iterations produce a mostly-complete paper. But it would take many, many iterations to pass all tests. In fact, I declared victory at iteration 36, since only two minor tests were failing. Appendix B provides more details, as does the ralph/run-final branch on Github.

The Paper Specification

paper-spec.md is the paper in bullet point form. For example, item I.3 lays out the arguments:

3. The paper makes the following arguments:
    a. Main argument: AI stocks may have high valuations, in part, because they help hedge against a negative AI singularity
    b. Incomplete markets are key: if markets are complete, there would be no need to hedge
    c. Financial market solutions to AI disaster risk are under-discussed, though frictions can limit their effectiveness.
    d. If the singularity occurs, then market frictions can be overcome due to the abundance of resources.
    e. Incomplete markets may distort not only valuations, but also the development of AI.

If you believe that the final test results are sufficient, you don't need to read the paper at all. paper-spec.md is all you need.

I wonder if, in the future, we will be submitting our paper specs and test results to conferences and journals.

The 25 Tests

I was hoping for just a few tests. A test suite with only

- factcheck-theory
- factcheck-narrative
- factcheck-code
- factcheck-lit

would have been beautiful.

But there so many more ways for a paper to be ugly. For example, I found that factcheck-anaphora was required to ensure the claims that are proved by reference are actually proved in the references.

I also found that having many tests is a good design. Like Ralph Wiggum, the AIs are easily distracted, and breaking up tests into smaller tasks helps them focus.

Comparison with the First Draft (April 2025)

12 months ago, the paper was generated by a sequence of LLM chat prompts:

```
Algo April 2025:
----------------
- LLM Prompt  1: draft the model description based on human-specified
                 functional forms and motivations
- LLM Prompt  2: find the price/dividend ratio
- LLM Prompt  3: generate a table using human-selected parameter values
...
```

Basically, I held AI's hand as we walked step-by-step through writing the paper. It was absurdly impractical.

Back then, the question was: did AI even write the paper? My conclusion was that, if they were human, I would have to give them credit as co-authors. After all, they did the math, writing, and lit reviews.

12 months later, this has entirely changed. Now, AI is so useful that it seems like there's no need to credit AI at all. Crediting an AI would be like crediting a statistical software package. Most readers should assume that AI were used throughout the research process, and would not be interested in which model was used where.

But some things haven't changed. 12 months ago, I complained that AI has sub-par economic reasoning:

> For example, Sonnet [3.7] often fails to recognize that an economic model does not capture an important channel. This is common (no model can capture everything). The standard practice is to dance gingerly around the channel in the writing. A decent PhD student can recognize this. But Sonnet cannot.

This is still true, even with Claude Opus 4.6.

What has changed is that we now can easily automate AI. It's easy to have an AI test for cautious discussions of unmodeled channels, and to have the other AIs keep working until the test passes. These improvements may not give AI the economic theory abilities of a decent Ph.D. student, but they do give them the persistence of Ralph Wiggum.

Limitations of the Current AI (April 2025)

Right now, AI is like a cognitive power tool, or cognitive heavy equipment. It can plow forward on just about any knowledge task. But it's difficult to wield it with subtlety or artistry.

The artisanal elements of research, like

- making connections across distant literatures
- turning dry economics into compelling narratives
- responding to referees and editors

these are still best done with humans in the loop. I might even add citing literature to this list. The element-lit-review test failed to cite some critical papers, including Korinek and Suh (2024); Zhang (2019, JF); and Babina et al. (2024).

I was hoping otherwise. I had an R&R in hand, and my dream was to have the AIs respond to the referee and editor. I attempted this by adding a `referee-cfr-r1` agent to the Ralph loop:

```
You are a constructive referee at a top finance journal.

You do NOT accept or reject this paper. You provide suggestions to help
the author improve it.

A prior referee (CFR R1) raised two concerns about an earlier draft.
That report is at `spec/CFR-R1-report`.
```

`referee-cfr-r1` produced decent suggestions, but the author agents struggled to

assemble the advice into a convincing response. The resulting extensions had inconsistent assumptions, theoretical dead ends, and weak economic takeaways.

Ultimately, I hard-coded the extension details. The final paper-spec.md specifies both the key modeling elements as well as the narrative takeaways.

Even with this hard coding, I found the Ralph loop's Extension 2 lacked finesse. I would have used a nonlinear deadweight cost to emphasize the drama of the singularity. The writing lacks the pull and rhythm that I strive for. I tried using writing-flow and similar tests to generate the pull but it was a disaster. The writing quality tests would constantly get into fights with element-gkp-cites, theory-clarity, and other tests. Ultimately, I deemed that automating finesse is out of the reach of current AI.

It was both disappointing and relieving.

Part of these struggles come from writing an unusual theory paper. The AIs did much better in my Ralph loop for a straightforward empirical paper. It's likely that there's much more empirical work in the training data. But additionally, I think theory exemplifies the subtle, artisanal aspects of research.

Overall, ralph-wiggum-asset-pricing generates a paper that is good enough for me to put my name on it, but not much more. To be honest, it's not my proudest work. I hope this preface helps make up for it.

The Future of AI and Economics Research (Speculative)

A common response to Novy-Marx and Velikov (2026): was "people are not ready for this." I heard concerns that peer review will be inundated with AI-generated slop.

Working on this paper gave me a different perspective. It made me think about the drivers of demand for research. I think the drivers are:

1. Readers want to learn something interesting and true.
2. Readers don't want to check all the work.
3. A system of author reputations makes 1 and 2 possible.

The current AI does not fundamentally alter these drivers. Critically, driver 3 made me quite wary of putting my name on AI slop. As a result, I don't see the journals becoming overwelmed by AI-generated submissions.

Instead, I see AI pushing the frontier of what is "interesting" and "true". Questions that can be answered by asking AI, or even by running a Ralph loop for a few days---these will soon be uninteresting. To get in the top journals---to

be declared "interesting" by the reputation system, that will require doing more. Accordingly, it will become harder to verify a paper is "true".

As a result, even if I could crank out papers that would have been publishable in 2025 with Ralph loops---I won't. I have a reputation to maintain!

I worry a bit that AI will push what is "interesting" so far that most peer-reviewed research becomes unintelligible and/or irrelevant. But I fear that we've already reached this point, without the help of AI.

My main worry is how AI affects the supply of research:

1. Economists use AI and labor to produce research of a certain quality.
2. Reputable authors use their attention to combine economists' research into papers that are read.

My base case is that AI *complements* economists' labor. As demonstrated by this repo, LLMs still struggle with the artisanal aspects of economics research, and I don't see this changing unless there's a paradigm shift. So in my base case, the number of economists remains the same, but our research output increases in terms of both quantity and quality.

But there are reasons why total research output is limited. Readers can only pay attention to so many reputable authors. These authors, in turn, can only pay attention to so many projects.

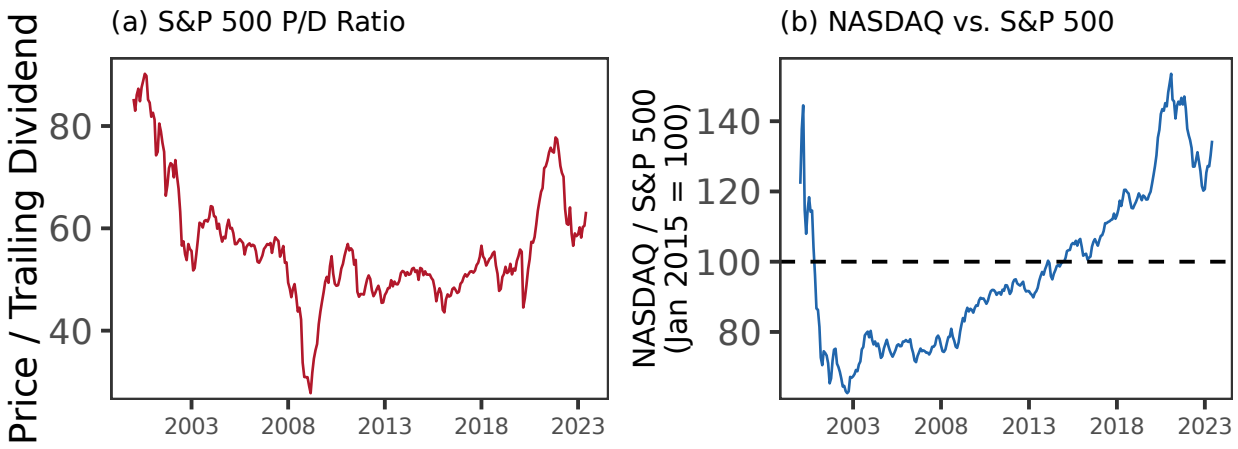
And of course, I might find myself shocked by AI progress, just as I was shocked 12 months ago. If AI progresses so much that certain models *become* the reputable authors, I shudder to think about the resulting job market.

To be clear, I don't *expect* a disaster. But even if it's highly unlikely, it's still a scenario that economists should consider.

1 Introduction

The publicly traded stocks most exposed to artificial intelligence have reached remarkable valuations. Figure 1 illustrates: the S&P 500's price-dividend ratio has reached historically elevated levels, and the AI- and technology-heavy NASDAQ Composite has sharply outpaced the broader market since 2015, with the valuation gap widening since 2023 as advances in generative AI have accelerated expectations of transformative productivity gains—and, implicitly, of the displacement such gains may bring.

Figure 1: Valuation Ratios and the AI Premium. Panel (a) shows the S&P 500 trailing price-dividend ratio from the Shiller dataset, which has reached historically elevated levels. Panel (b) shows the NASDAQ Composite price relative to the S&P 500, normalized to January 2015 = 100; the NASDAQ is heavily weighted toward AI and technology firms, so a rising ratio indicates growing relative valuations. Sources: NASDAQ from FRED; S&P 500 from the Shiller dataset.



Part of this premium, we argue, reflects a hedging motive: investors use AI stocks to partially insure against displacement from AI. We define a *negative* AI singularity as a sudden, dramatic improvement in AI productivity that displaces the typical investor's labor income and consumption. Much of the relevant AI equity is restricted—held by founders, early-stage investors, and firms that may not yet exist. As Gârleanu, Kogan, and Panageas (2012) emphasize, the displacing capital often belongs to future innovators who have not yet entered the economy, so investors cannot trade it. Because investors cannot trade the restricted equity, they turn to publicly traded AI stocks as the only available partial hedge against displacement, pushing valuations above those of non-AI stocks—a premium that would largely vanish if markets were complete.

To formalize this mechanism, we build on Gârleanu, Kogan, and Panageas (2012)'s framework by modeling a discrete AI singularity with severe displacement. A representative household—the marginal investor—faces a stochastic singularity that shifts consumption toward AI owners holding private capital. AI stocks grow as a share of the economy with each singularity, making them a partial hedge. The model delivers closed-form price-dividend ratios, and quantitative analysis shows that at a singularity probability of one percent, P/D ratios for AI stocks roughly double relative to non-AI stocks, consistent with observed valuation spreads. Extinction risk (Jones, 2024) partially offsets this premium: the states in which AI is powerful enough to produce enormous growth are also those in which existential risk is highest, narrowing the valuation spread (Proposition 2). But while extinction risk at-

tenuates the pricing distortion, market incompleteness itself has consequences beyond asset prices.

The same incompleteness that inflates AI valuations also distorts real decisions—it can distort the development of AI itself. When the positive singularity is more likely than the negative one, AI development is socially efficient, yet a risk-averse household that cannot hedge displacement may rationally choose to block it—the uninsurable downside looms larger than the expected upside (Proposition 3). Calls to slow or halt AI development may partly reflect investors' inability to hedge the downside.

Financial approaches to AI disaster risk are strikingly under-discussed relative to regulatory and alignment-focused approaches, and frictions severely limit the solutions that do exist. The natural fix—broader trading of AI equity—is blocked by restricted ownership and non-existent future capital. Government transfers can substitute for missing markets, but standard fiscal tools are limited by deadweight costs that scale with the size of transfers, making them ineffective in ordinary settings.

The singularity setting, however, offers a distinctive resolution. If the singularity occurs, these frictions can be overcome due to the sheer abundance of resources. If a singularity produces the kind of explosive output growth modeled by Jones (2024), even grossly inefficient transfers become effective, because the resource base overwhelms the deadweight costs. The same explosive growth that drives the incomplete-markets problem also provides the means to overcome it through redistribution.

We develop these ideas as follows. Section 2 presents the model and derives equilibrium prices, Section 3 provides quantitative analysis, and Section 4 develops extensions on AI development distortions and government transfers.[1]

Related literature. Our paper builds most directly on Gârleanu, Kogan, and Panageas (2012), who model how innovation displaces existing agents and creates a systematic risk factor under incomplete markets. The main insights about displacement risk and incomplete markets originate in their work; we connect these ideas to the AI singularity and examine how explosive growth interacts with government transfers.

Jones (2024) studies the trade-off between AI-driven growth and existential risk. We incorporate his extinction channel and show it attenuates rather than amplifies the valuation premium. Our work also relates to creative destruction and displacement risk premia (Kogan and Papanikolaou, 2014; Kogan, Papanikolaou, and Stoffman, 2020; Knesl, 2023), the macroeconomics of AI growth (Aghion, Jones, and Jones, 2019; Acemoglu, 2025), the rare disasters literature (Barro, 2006; Wachter, 2013), and Pástor and Veronesi (2009)'s analysis

[1] This paper is itself a product of the displacement it models. The analysis, code, and prose were produced by AI agents. See `https://github.com/redacted/ralph-wiggum-asset-pricing/`

of technological revolutions.

2 Model

2.1 Setup

Time is discrete and infinite, $t = 0, 1, 2, \ldots$ A representative household is the marginal investor in public stock markets. There is also a group of AI owners who hold private AI capital and are not marginal investors in public stocks. The AI owners can also be thought of as future capital owners who do not yet participate in markets, as in Gârleanu, Kogan, and Panageas (2012). Importantly, we do not explicitly model the entry of new cohorts of firms or workers; AI owners are a static group whose share changes only through the singularity mechanism.

Consumption. Aggregate consumption at date t is C_t. In the absence of a singularity, aggregate consumption grows at a constant rate $g > 0$:

$$C_{t+1} = (1 + g)\, C_t. \tag{1}$$

The household receives a share $\alpha_t \in (0, 1)$ of aggregate consumption, so its consumption is $c_t^H = \alpha_t\, C_t$. AI owners receive the remainder, $(1 - \alpha_t)\, C_t$.

Singularity. Each period, with probability p, an AI singularity occurs. With probability $1 - p$, no singularity occurs and α_t is unchanged. Conditional on a singularity:

1. With probability $1-\xi$, the singularity is *non-extinction*. Aggregate consumption jumps by a factor $1 + \eta$ (where $\eta > 0$ captures the productivity boost), but the household's share drops:

$$\alpha_{t+1} = \phi\, \alpha_t, \qquad \phi \in (0, 1). \tag{2}$$

   The parameter ϕ governs displacement severity: smaller ϕ means larger displacement.

2. With probability ξ, the singularity triggers *extinction*: $C_{t+1} = 0$ for all subsequent dates. This follows Jones (2024), who emphasizes that the states in which AI is powerful enough to produce enormous growth are precisely those in which existential risk is highest.

After a non-extinction singularity, the economy continues with the new, lower household share. Singularities can occur repeatedly, progressively displacing the household.

Assets. Two public assets are available for trading:

- **AI stocks** pay dividends $D_t^{AI} = \theta_t C_t$, where θ_t is the AI dividend share. Upon a non-extinction singularity, the AI share expands:

$$\theta_{t+1} = \theta_t + \Delta\theta\,(1 - \theta_t), \qquad \Delta\theta \in (0, 1). \tag{3}$$

  This is the expression that drives $\Gamma^{AI} \neq \Gamma^N$ in equilibrium: AI stocks' dividends grow faster than the economy upon a singularity, while non-AI stocks shrink as a share. Absent a singularity, θ_t is unchanged.

- **Non-AI stocks** pay dividends $D_t^N = (1 - \theta_t)\,C_t$.

Total public dividends equal aggregate consumption ($D^{AI} + D^N = C_t$), but these dividends are distributed among all investors, not solely the household. The household's consumption $\alpha_t C_t$ reflects the net outcome of its portfolio returns from public stocks and any labor-like income; AI owners receive the remainder through their own public stockholdings and their restricted equity. Because α_t and θ_t are distinct—one governs the consumption split, the other the dividend split among publicly traded assets—the household's share can differ from its pro-rata claim on public dividends.

AI owners also hold restricted equity—founder stakes, pre-IPO shares, and other claims on AI firms that are not available for public trading. The household *cannot* purchase these restricted shares. This is the source of market incompleteness: although the household can trade publicly listed AI stocks, it cannot access the full universe of AI equity, and therefore cannot fully hedge displacement risk.

The displacement parameter ϕ governs the household's total consumption share, which reflects both tradeable and non-tradeable components (labor income, private wealth, and returns from public stockholdings). Holding AI stocks allows the household to smooth marginal utility across states—this is the hedging channel that inflates AI stock valuations—but does not eliminate displacement, because the non-tradeable component (primarily labor income) is what ϕ captures. If the household could trade the restricted equity, it could hedge this component directly, and the valuation spread would collapse.

Preferences. The household has CRRA preferences with risk aversion $\gamma > 1$ and discount factor $\beta \in (0, 1)$:

$$U_0^H = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \, \frac{(c_t^H)^{1-\gamma}}{1-\gamma} \right]. \tag{4}$$

2.2 Equilibrium prices

The household prices all publicly traded assets via its Euler equation. Because markets are incomplete, the household's stochastic discount factor (SDF) reflects its own consumption growth, not aggregate consumption growth.

Proposition 1 (Price-dividend ratios)**.** *In the stationary equilibrium where the household holds both public assets, the price-dividend ratios are:*

$$\frac{P^{AI}}{D^{AI}} = \frac{\beta(1+g)^{1-\gamma}\left[(1-p)+p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{AI}\right]}{1-\beta(1+g)^{1-\gamma}\left[(1-p)+p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{AI}\right]}, \tag{5}$$

$$\frac{P^{N}}{D^{N}} = \frac{\beta(1+g)^{1-\gamma}\left[(1-p)+p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{N}\right]}{1-\beta(1+g)^{1-\gamma}\left[(1-p)+p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{N}\right]}, \tag{6}$$

where $\Gamma^{AI} = \frac{\theta+\Delta\theta(1-\theta)}{\theta}(1+\eta)$ *and* $\Gamma^{N} = \frac{1-\theta-\Delta\theta(1-\theta)}{1-\theta}(1+\eta)$ *are the dividend growth factors for AI and non-AI stocks conditional on a non-extinction singularity.*

Proof. See Appendix A. □

Remark 1 (Existence condition)**.** *The P/D ratios in Proposition 1 are well-defined (positive and finite) if and only if the numerator*

$$A^{j} \;\equiv\; \beta(1+g)^{1-\gamma}\left[(1-p)+p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\,\Gamma^{j}\right] < 1. \tag{7}$$

When $A^j \geq 1$*, the SDF-weighted expected dividend growth exceeds the discount rate and the geometric pricing sum diverges—the asset's price is infinite. Intuitively, the hedging value of the asset is so extreme that no finite price can clear the market. The baseline calibration satisfies* $A^j < 1$ *for both assets. As we discuss in Section 4.2, sufficiently severe displacement can violate this condition, which itself motivates the role of government transfers.*

The closed-form expressions in Proposition 1 rely on an approximation: the post-singularity P/D ratio is treated as equal to the pre-singularity ratio. This is exact when $\Delta\theta \to 0$ and becomes less accurate as $\Delta\theta$ grows, since each singularity shifts θ and therefore shifts the dividend growth factors. The closed-form is useful for building intuition about the hedging channel, particularly through the comparison of Γ^{AI} and Γ^{N}. Table 1 reports numerically exact P/D ratios, computed by iterating the Euler equation over the chain of post-singularity states (see Appendix A). The full derivation of the closed form is also in Appendix A.

The key economic content of Proposition 1 is in the comparison of Γ^{AI} and Γ^{N}. Since $\Delta\theta > 0$, AI stocks' dividends grow faster than aggregate consumption upon a singularity ($\Gamma^{AI} > 1+\eta$), while non-AI stocks' dividends grow more slowly ($\Gamma^{N} < 1+\eta$), shrinking

as a share of the economy. When $\phi(1+\eta) < 1$—that is, when displacement is severe enough that the household's consumption falls despite the aggregate productivity gain—the household's marginal utility in singularity states is high, and AI stocks' payoffs are especially valuable, pushing their P/D ratio above that of non-AI stocks. This is the hedging channel: AI stocks pay off precisely when the household's consumption falls, making them a partial hedge against displacement.

It is immediate from the P/D expressions that the valuation spread widens with displacement severity (decreasing ϕ, which raises marginal utility in singularity states via $\phi^{-\gamma}$) and with singularity probability p for sufficiently risk-averse households (higher p puts more weight on the states where AI stocks' dividend advantage operates). The interaction with extinction risk is less obvious:

Proposition 2 (Extinction attenuation)**.** *The valuation spread is decreasing in extinction probability ξ.*

Proof. Higher ξ reduces the weight on non-extinction singularity states via the factor $(1-\xi)$ in each P/D ratio. Since these are the only states where AI stocks' dividend advantage ($\Gamma^{AI} > \Gamma^{N}$) operates, higher extinction probability compresses the valuation spread. □

2.3 Discussion

The model's mechanism parallels Gârleanu, Kogan, and Panageas (2012), who show that growth stocks earn lower expected returns because they hedge displacement risk from innovation. In their framework, displacement is driven by new cohorts of firms entering the economy; in ours, it is driven by a discrete AI singularity. The key difference is the nature of the displacement event: GKP model continuous displacement from expanding technological variety, while we model a sudden, severe shift in which the household's consumption falls even as aggregate output rises. This makes the interaction with extinction risk (Jones, 2024) natural and generates sharper quantitative predictions about how singularity probabilities map to valuation ratios.

The market incompleteness in our model—the household's inability to trade restricted AI equity—is central. If the household could trade the restricted equity, its effective displacement parameter would become $\phi_{\text{eff}} \to 1$: displacement is fully hedged, the SDF no longer overweights singularity states, and the displacement-driven valuation premium largely collapses—though a small residual spread from differential dividend growth ($\Gamma^{AI} \neq \Gamma^{N}$) remains. This echoes Gârleanu, Kogan, and Panageas (2012)'s point that future innovators' rents cannot be traded because the innovators have not yet entered the economy. Our AI owners play an analogous role, though we do not model the entry dynamics that are central

to their framework; the displacement in our model comes from the singularity's reallocation of consumption shares rather than from creative destruction by new entrants.

One feature of the discrete singularity that has no analog in Gârleanu, Kogan, and Panageas (2012)'s continuous-displacement framework is that sufficiently severe displacement can violate the existence condition for finite prices (Remark 1). When displacement is extreme, the household's marginal utility in singularity states becomes so large that the SDF-weighted expected dividend growth exceeds the discount rate, and no finite price can clear the market. This discontinuity—from finite to infinite hedging demand—cannot arise under GKP's gradual displacement, where the pricing kernel remains well-behaved. It reflects the distinctive severity of a discrete AI singularity and, as we show in the extensions, motivates the role of government transfers.

3 Quantitative Analysis

Table 1 reports price-dividend ratios across a grid of singularity probabilities and extinction risks. The parameterization uses $\beta = 0.96$, $g = 0.02$, $\gamma = 4$, $\phi = 0.5$ (the household retains half of its consumption share upon displacement, so that $\phi(1 + \eta) = 0.75$ and household consumption falls by 25%), $\eta = 0.5$ (aggregate consumption rises by 50% in a singularity), $\theta = 0.15$ (AI stocks are initially 15% of the economy), and $\Delta\theta = 0.2$ (AI's share jumps by 20% of the non-AI remainder upon singularity).

Several patterns emerge. First, AI stock P/D ratios are substantially higher than non-AI stock P/D ratios across the entire grid. For a singularity probability of 0.5% per year with no extinction risk, AI stocks trade at a P/D of about 15.5, while non-AI stocks trade near 11—a ratio of about 1.4. At $p = 1\%$, the ratio rises to 2. Second, increasing the singularity probability raises the AI stock premium: the household values the hedge more as the risk it hedges against becomes more likely. Third, extinction risk compresses the AI premium, as Proposition 2 predicts. At high extinction probabilities, even AI stocks lose value because the states in which they pay off become less likely, reducing both valuations and narrowing the spread.

The magnitudes are broadly suggestive. As Figure 1 shows, the S&P 500 P/D ratio has reached historically elevated levels, and the AI-heavy NASDAQ has appreciated roughly 50% more than the S&P 500 since 2015, consistent with rising relative valuations for AI-exposed stocks. The mapping from NASDAQ vs. S&P 500 to the model's AI vs. non-AI distinction is imperfect: the NASDAQ is broader than "AI stocks," return differences partly reflect earnings growth rather than valuation multiples, and the S&P 500 itself has substantial AI exposure. Nonetheless, the pattern is consistent with a sustained valuation premium of the

Table 1: Price-Dividend Ratios: AI Stocks vs. Non-AI Stocks

		Price-Dividend Ratio		
p	ξ	AI Stocks	Non-AI Stocks	Ratio
0.1%	0%	10.4	9.8	1.1
0.1%	5%	10.4	9.8	1.1
0.1%	10%	10.3	9.7	1.1
0.1%	20%	10.2	9.7	1.1
0.2%	0%	11.5	10.1	1.1
0.2%	5%	11.4	10.0	1.1
0.2%	10%	11.3	10.0	1.1
0.2%	20%	11.0	9.9	1.1
0.5%	0%	15.5	11.1	1.4
0.5%	5%	15.0	11.0	1.4
0.5%	10%	14.6	10.8	1.3
0.5%	20%	13.8	10.6	1.3
0.8%	0%	21.2	12.3	1.7
0.8%	5%	20.2	12.1	1.7
0.8%	10%	19.2	11.8	1.6
0.8%	20%	17.4	11.4	1.5
1.0%	0%	26.5	13.3	2.0
1.0%	5%	24.8	12.9	1.9
1.0%	10%	23.2	12.6	1.8
1.0%	20%	20.5	12.0	1.7

Parameters: $\beta = 0.96$, $g = 0.02$, $\gamma = 4$, $\phi = 0.5$, $\eta = 0.5$, $\theta = 0.15$, $\Delta\theta = 0.2$. p is the annual singularity probability; ξ is the extinction probability conditional on singularity. AI P/D ratios are numerically exact, computed by iterating the Euler equation over post-singularity states (see Appendix A). The ratio column reports $\mathrm{P/D}^{AI}/\mathrm{P/D}^{N}$.

kind the model predicts, where AI stock P/D ratios are 1.3 to 2 times those of non-AI stocks across annual singularity probabilities in the 0.5–1% range.

4 Extensions: Market Incompleteness and the Singularity

The baseline model takes market incompleteness as given and studies its pricing implications. This section examines two further consequences: how incompleteness distorts the development of AI, and how government policy might address it. To simplify notation, we write α for the household's current consumption share α_t.

4.1 Extension 1: Veto and efficient development

We augment the baseline model to allow a positive singularity. With probability p, a singularity occurs. Conditional on a non-extinction singularity (probability $1-\xi$), the singularity is either *positive* (probability q)—the household's share increases to $\alpha_{t+1} = \min(1, \alpha_t/\phi)$ and aggregate consumption jumps by $1+\eta$—or *negative* (probability $1-q$, as in the baseline), with the household's share falling to $\alpha_{t+1} = \phi\,\alpha_t$. We assume $q > 1/2$: the positive singularity is the more likely outcome.

We say AI development is *socially efficient in the Kaldor-Hicks sense*: the total surplus from a non-extinction singularity—summing over both the household and AI owners—is positive, so that the winners could in principle compensate the losers. This holds when $(1+\eta) > 1$, since aggregate consumption rises by a factor $1+\eta$ in both singularity outcomes, and the only question is how the gains are distributed.

The household can *veto* AI development at a cost $\kappa > 0$, representing a permanent fraction of consumption lost to the deadweight costs of intense government intervention needed to halt AI progress. If the household vetoes, it consumes $(1-\kappa)\alpha C_t$ each period with no singularity. We normalize extinction utility to zero; since CRRA utility is negative for all $c > 0$ when $\gamma > 1$, this makes the veto result conservative.

Under complete markets, the household can trade the restricted AI equity—founder stakes, pre-IPO shares, and other claims currently unavailable—so that it fully hedges displacement risk. Its consumption in both singularity states then equals its pre-singularity share of aggregate consumption: $\alpha(1+\eta)C_t(1+g)$.

Proposition 3 (Veto under incomplete markets)**.** *(i) Under incomplete markets, suppose $\phi(1+\eta) < 1$ (the household's consumption falls upon a negative singularity). Then there exists a threshold $\bar{\gamma}$ such that for all $\gamma > \bar{\gamma}$, the household vetoes AI development ($V_{veto} > V_{develop}$), even when development is socially efficient and the veto cost κ is positive.*

(ii) Under complete markets: for κ sufficiently small (in particular, for the same κ as in part (i)), the household never vetoes socially efficient AI development ($V^{CM}_{develop} > V_{veto}$).

Proof. (i) The household's expected one-period utility gain from a non-extinction singularity under incomplete markets is

$$\Delta u(\gamma) = q\,u(\alpha^{+}(1+\eta)) + (1-q)\,u(\phi\alpha(1+\eta)) - u(\alpha), \tag{8}$$

where $\alpha^{+} = \min(1, \alpha/\phi)$ and $u(c) = c^{1-\gamma}/(1-\gamma)$. As $\gamma \to \infty$, the negative-singularity term dominates because $\phi\alpha(1+\eta) < \alpha$ when $\phi(1+\eta) < 1$: the utility cost of the consumption

drop grows without bound relative to the utility gain from the positive singularity. Formally, $\lim_{\gamma\to\infty} \Delta u(\gamma)/u(\alpha) = +\infty$ when $\phi(1+\eta) < 1$. In the infinite-horizon problem, $V_{\text{develop}} - V_{\text{veto}}$ is a discounted sum of per-period utility differences; the singularity term contributes $p(1-\xi)\Delta u(\gamma)$ in expectation each period, and since $\Delta u(\gamma) \to -\infty$ while the veto cost κ is fixed, the infinite-horizon difference diverges to $-\infty$ for γ large enough.

(ii) Under complete markets, the household's consumption in both singularity states is $\alpha(1+\eta)C_t(1+g)$. Its expected utility gain is $u(\alpha(1+\eta)) - u(\alpha)$, which is positive since $\eta > 0$. The household strictly prefers development for any $\gamma > 1$ and any κ small enough, and in particular for the same κ used in part (i). □

The veto result illustrates how market incompleteness can distort real decisions, not just asset prices. A numerical example sharpens the point. Using the baseline displacement parameters ($\phi = 0.5$, $\eta = 0.5$, $\xi = 5\%$) with $p = 1\%$, $\gamma = 10$, $\alpha = 0.70$, $q = 0.70$, and a veto cost $\kappa = 1\%$ of permanent consumption: the household vetoes under incomplete markets ($V_{\text{veto}} > V_{\text{develop}}$), even though the positive singularity is more than twice as likely as the negative one. The computation solves the infinite-horizon Bellman equation, treating the singularity as a one-shot event: each period the singularity occurs with probability p, and after realization the economy continues with deterministic growth at the new consumption share. Under complete markets with the same parameters, the household does not veto ($V^{CM}_{\text{develop}} > V_{\text{veto}}$), since it can trade claims on AI capital and consume $\alpha(1+\eta)C_t(1+g)$ in both singularity states. Allowing repeated singularities would reinforce the veto motive by compounding displacement risk.

Extinction risk interacts with this distortion: under our conservative normalization ($U_{\text{ext}} = 0$), higher ξ reduces the weight on non-extinction singularity states, which are the states driving the veto. With a more severe extinction penalty (as Jones (2024) argues is appropriate for bounded utility), higher ξ would amplify the veto incentive. This connects to debates about AI regulation: Proposition 3 implies that calls to slow or halt AI development may partly reflect unhedgeable downside risk from displacement—the veto is a rational response to the consumption loss that markets cannot insure, not merely technophobia.

Jones (2024) identifies two distinct channels through which preferences shape attitudes toward AI risk: agents with higher risk aversion γ are less willing to accept existential gambles, and agents with higher consumption levels value their current living standards more, independently reducing their tolerance for catastrophic outcomes. Our model offers a complementary channel through financial markets rather than existential risk. Under incomplete markets, households with larger consumption shares—more exposed to displacement—have stronger veto incentives, because the unhedgeable downside looms larger relative to the upside they cannot fully capture. This connects wealth heterogeneity to attitudes about AI

development through the displacement channel, not just through differential valuations of life.

4.2 Extension 2: Government transfers

Extension 1 showed that market incompleteness can cause the household to veto socially efficient AI development. A natural question is whether policy can restore efficiency. The ideal solution—broader trading of AI capital—faces the same constraint Gârleanu, Kogan, and Panageas (2012) identify: much of the displacing capital does not yet exist. Government transfers offer an alternative, and the singularity setting gives them a distinctive advantage: if the productivity jump is large enough, the abundance of post-singularity resources can overwhelm the deadweight costs that normally make transfers ineffective. Transfers thus serve a dual role: they reduce AI stock valuations by cushioning displacement (a pricing effect) and, by attenuating the household's unhedgeable downside, they can eliminate the veto distortion (a real effect).

Gârleanu, Kogan, and Panageas (2012) note, in the course of establishing robustness, that intergenerational transfers mandated by the government would affect the magnitude of the displacement factor, listing this alongside bequests, gifts, government debt, and adjustable capital as extensions that preserve the functional form of their key pricing equation. We take this further by analyzing how government transfers interact with displacement in the specific setting of an AI singularity, where the key question is whether explosive output growth can overcome the deadweight costs that normally make transfers ineffective. Because the displacing equity may not yet exist—it belongs to future cohorts of innovators—direct trading is infeasible, and government transfers offer an alternative. But transfers ordinarily incur deadweight costs (waste, fraud, administrative burden) that scale with transfer size, making them unattractive.

We model this as follows. The government imposes a tax rate $\tau \in [0, 1)$ on AI owners' post-singularity consumption and transfers the proceeds to the household. Deadweight costs consume a fraction $\delta\tau$ of the transferred amount, where $\delta > 0$ governs the severity of waste. The household's post-transfer consumption in a non-extinction singularity state is:

$$c^H_{post} = \phi\,\alpha\,(1+\eta)\,C_t\,(1+g) + \tau\,(1-\delta\tau)\,(1-\phi\alpha)\,(1+\eta)\,C_t\,(1+g). \tag{9}$$

The first term is the household's displaced consumption. The second is the net transfer: a fraction τ of the AI surplus, reduced by the deadweight cost $\delta\tau$. Note that $(1-\phi\alpha)$ represents the AI owners' share of post-singularity aggregate consumption; this expression uses the household's consumption share α rather than the AI dividend share θ, because the

transfer is levied on the consumption allocation, not on publicly traded dividends.

To compute AI stock valuations under transfers, factor equation (9) as $c^H_{post} = \phi_{\text{eff}} \cdot \alpha \cdot (1+\eta)(1+g)C_t$, where

$$\phi_{\text{eff}} \;=\; \phi + \frac{\tau(1-\delta\tau)(1-\phi\alpha)}{\alpha} \tag{10}$$

is the effective displacement parameter. Equation (10) follows directly from dividing (9) by $\alpha(1+\eta)(1+g)C_t$: the first term contributes ϕ and the transfer term contributes the remainder. Since ϕ_{eff} enters the SDF in the same way as ϕ, the P/D formula from Proposition 1 applies with ϕ replaced by ϕ_{eff}. For simplicity, we compute ϕ_{eff} at the initial share α_0; after successive singularities, α declines, slightly altering the effective transfer. This approximation does not affect the qualitative conclusions.

In standard settings (moderate η), the deadweight costs eat into the transfer and the household gains little. But in a singularity with large η, aggregate output grows enormously. The transfer base—AI owners' surplus—grows with η, while the deadweight cost rate is fixed. The ratio of post-transfer to pre-transfer household consumption in the singularity state is independent of the productivity jump η:

$$\frac{c^H_{post}}{c^H_{no\text{-}transfer}} = 1 + \frac{\tau(1-\delta\tau)(1-\phi\alpha)}{\phi\alpha}. \tag{11}$$

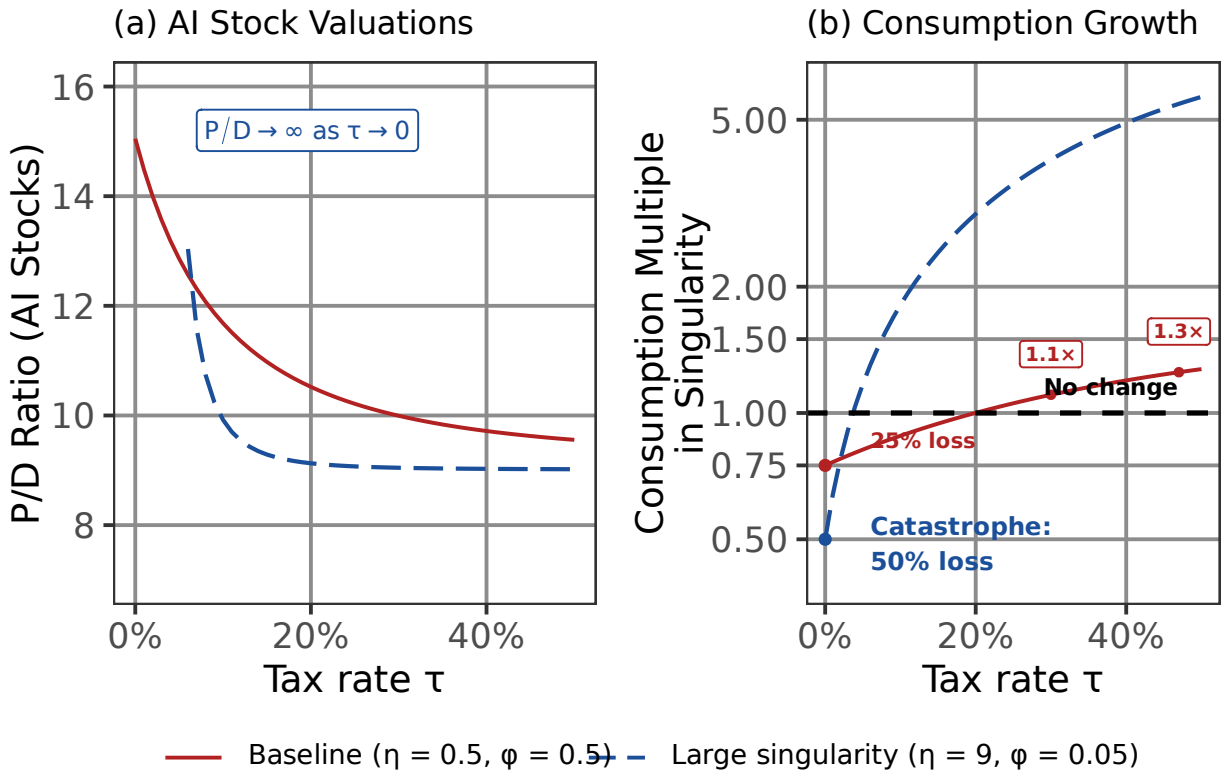
This ratio exceeds one whenever $\tau > 0$ and $\delta\tau < 1$ (i.e., deadweight costs do not consume the entire transfer): transfers always improve the household's position in the singularity state, regardless of η. The economic content is in the *levels*: as η grows, both c^H_{post} and $c^H_{no\text{-}transfer}$ grow without bound, so even inefficient transfers deliver arbitrarily large consumption gains relative to the pre-singularity baseline. To illustrate the robustness to even more severe waste: under the large singularity ($\eta = 9$, $\phi = 0.05$), raising the deadweight cost parameter to $\delta = 0.9$ (compared with the $\delta = 0.5$ used in Figure 2)—so that net transfers amount to only $\tau(1-\delta\tau)$ of the AI surplus, e.g., 0.219 at $\tau = 0.30$—still yields a consumption multiple of 3.5× at $\tau = 0.30$, compared to the 0.5× catastrophe without transfers. Even grossly inefficient redistribution transforms a 50% consumption loss into a 250% gain when the resource base is large enough.

Figure 2 illustrates with two panels, using the baseline risk aversion $\gamma = 4$, $\alpha = 0.70$, $p = 0.5\%$, and $\xi = 5\%$. The baseline uses $\eta = 0.5$ and $\phi = 0.5$; the large singularity uses $\eta = 9$ with $\phi = 0.05$, reflecting the intuition that a singularity powerful enough to produce ten-fold growth would also displace far more labor. The right panel shows that absent transfers ($\tau = 0$), the household faces a catastrophe: consumption halves under the large singularity ($\phi(1+\eta) = 0.5$) and falls by 25% under the baseline ($\phi(1+\eta) = 0.75$). As τ

increases, the large singularity responds dramatically—the enormous output growth swamps the deadweight costs—while the baseline gains only modestly.

The left panel shows the corresponding effect on AI stock valuations: transfers reduce the hedge value of AI stocks, compressing P/D ratios. Under the large-singularity parameters, the P/D ratio is undefined at $\tau = 0$: the existence condition in Remark 1 is violated because the household's marginal utility in the singularity state ($\phi^{-\gamma} = 160{,}000$) is so extreme that the pricing sum diverges. In economic terms, the hedge value of AI stocks becomes infinite—no finite price can compensate for the catastrophic displacement. As τ increases and transfers cushion the displacement, effective displacement falls, the existence condition is restored, and finite P/D ratios emerge. The transition from infinite to finite prices illustrates the severity of the incomplete-markets problem and the potential value of transfers.

Figure 2: Government Transfers and the Singularity. Panel (a) shows how transfers compress AI stock P/D ratios by reducing the household's hedging demand. Under the large-singularity parameterization, the P/D ratio is undefined at low tax rates because the existence condition (Remark 1) is violated: displacement is so severe that the household's marginal utility makes the pricing sum diverge, and no finite price can clear the market. Transfers restore finite prices by cushioning displacement. Panel (b) shows the household's consumption change in the singularity state: absent transfers, the household faces a catastrophe (marked at $\tau = 0$), but under the large singularity, even modest tax rates produce enormous consumption gains as explosive output growth overwhelms deadweight costs. Parameters: $\alpha = 0.70$, $p = 0.5\%$, $\xi = 5\%$, $\delta = 0.5$.



The policy implication is nuanced. In normal times, government transfers are a blunt and wasteful tool for addressing displacement risk. But if an AI singularity produces the kind of explosive output growth modeled by Jones (2024) and examined critically by Nordhaus (2021), the calculus changes: the abundance of resources makes even inefficient redistribution effective. This suggests that contingent transfer policies—triggered by a singularity—may be worth designing in advance.

5 Conclusion

AI stocks trade at high valuations. We have argued that part of this premium reflects a hedging motive: investors use AI stocks to partially insure against an AI singularity that would displace their consumption. This mechanism requires market incompleteness—the inability to trade restricted AI equity—and is attenuated by extinction risk, which reduces the weight on the non-extinction states where the hedging channel operates. When markets are incomplete, risk-averse households may inefficiently block AI development, and government transfers, normally wasteful, can become effective if singularity-driven growth is large enough to overwhelm deadweight costs.

Our model is deliberately simple. It abstracts from continuous-time dynamics, heterogeneous beliefs, production-side details, and many other features that would enrich the analysis. The goal is not to provide a definitive account of AI stock valuations but to highlight a specific channel—hedging against displacement under incomplete markets—that connects asset pricing theory to the distinctive features of the AI singularity.

A Proof of Proposition 1

The household's Euler equation for any asset j with price P_t^j and dividend D_t^j is:

$$P_t^j = \mathbb{E}_t \left[\beta \left(\frac{c_{t+1}^H}{c_t^H} \right)^{-\gamma} (P_{t+1}^j + D_{t+1}^j) \right]. \tag{12}$$

Consider the AI stock. Write the price-dividend ratio as $v^{AI} = P^{AI}/D^{AI}$, which is constant in the stationary equilibrium before any singularity has occurred. The household's consumption growth c_{t+1}^H/c_t^H takes three values:

- No singularity (prob $1-p$): $c_{t+1}^H/c_t^H = 1+g$, and $D_{t+1}^{AI}/D_t^{AI} = 1+g$.
- Non-extinction singularity (prob $p(1-\xi)$): $c_{t+1}^H/c_t^H = \phi(1+g)(1+\eta)$, and $D_{t+1}^{AI}/D_t^{AI} = \Gamma^{AI}(1+g)$.
- Extinction (prob $p\xi$): $c_{t+1}^H = 0$, so the payoff is zero.

Substituting into the Euler equation with $P_t^{AI} = v^{AI} D_t^{AI}$:

$$\begin{aligned} v^{AI} D_t^{AI} = \beta(1+g)^{-\gamma} \Big[& (1-p)(1+g)(v^{AI}+1) D_t^{AI} \\ & + p(1-\xi)[\phi(1+\eta)]^{-\gamma}(1+g)\, \Gamma^{AI}(v^{AI}+1) D_t^{AI} \Big]. \end{aligned} \tag{13}$$

Note that after a non-extinction singularity the economy resets with new parameters (α', θ'), and the *new* P/D ratio in the post-singularity state will generally differ because Γ^{AI} depends on θ. (The non-AI growth factor $\Gamma^N = (1 - \Delta\theta)(1 + \eta)$ is θ-independent, so the non-AI closed form is exact.) For tractability, we approximate the post-singularity AI P/D ratio by v^{AI}. This approximation becomes exact as $\Delta\theta \to 0$. For the numerically exact values reported in Table 1, we solve the Euler equation by backward recursion over the chain $\theta_0, \theta_1 = \theta_0 + \Delta\theta(1 - \theta_0), \theta_2, \ldots$, starting from a terminal state where θ is close to one and the approximation error vanishes. Dividing by D_t^{AI} and solving for v^{AI}:

$$v^{AI} = \frac{\beta(1+g)^{1-\gamma}\left[(1-p) + p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\Gamma^{AI}\right]}{1 - \beta(1+g)^{1-\gamma}\left[(1-p) + p(1-\xi)(1+\eta)^{-\gamma}\phi^{-\gamma}\Gamma^{AI}\right]}. \tag{14}$$

This can be rewritten as equation (5). The derivation for non-AI stocks is identical, replacing Γ^{AI} with Γ^N.

B Appendix: Ralph's Work by Iteration (Human-Written)

Full disclosure: this appendix is mostly human-written.

Before starting the loop, I used `check-ralph-direction.sh`, which runs only the author plan and improve steps of the loop, to generate `several paper drafts`. I picked the most promising draft and initialized the Ralph loop with it.

The `Ralph loop` runs on its own branch, and commits after each iteration. So you can see in painstaking detail how the paper progressed on `ralph/run-final`.

Usually, the very first `author-plan` and `author-improve` prompts generate a fairly complete paper. Then, the first 3–4 iterations catch the major issues. This was evident in the final run:

- `iter 1` fixed:
  - Missing core literature on displacement risk and asset pricing, especially Kogan–Papanikolaou line of work.
  - Opening figure uses fabricated data.
  - Inconsistencies between paper's displayed P/D formulas and code.
  - Abstract/body mismatch on extinction risk distorting AI development.
  - Model inconsistency in treatment of private AI capital dividends and unused risk-free-rate notation.
  - Abstract exceeding the 100-word limit.
  - Missing clarification that paper does not model GKP-style entry of future innovators.
- `iter 2` fixed:
  - Overstatement of what GKP say about broader trading solving displacement risk.
  - Introduction overusing AI-authorship device in a repetitive, awkward way.
  - Verbal claims around valuation spreads and Proposition 3 not matching the paper's own results.
  - Appendix/proof material being mostly ceremonial rather than doing real argumentative work.
- `iter 3` fixed:
  - Lit review exceeding half-page spec limit.

  - Wrong attribution of JPE 2020 displacement/inequality paper and inconsistent terminology in transfer section.
  - Long proof of Proposition 1 appearing inline instead of being moved to the appendix.
  - Figure 2 text being too small to read.
  - Colored hyperlink boxes showing up in compiled PDF.

- `iter 4` fixed:
  - Opening figure still using fabricated data.
  - Transfer figure not clearly showing that households are badly harmed with zero transfers.
  - Citation errors around Kogan-Papanikolaou-related authorship and overly favorable framing of Nordhaus.
  - Missing existence condition for Proposition 1, which made some Extension 2 valuations mathematically undefined.

After the `iter 4` fixes, the remaining test failures were:

- `element-gkp-cites`: Mischaracterization of GKP's mechanism, inaccurate/insensitive "inessential extension" language.
- `element-lit-review`: Missing Knesl from lit review.
- `factcheck-bysection`: Text says non-AI dividends "shrink" but they only shrink relative to the economy.
- `factcheck-freely`: Remaining bibliography problem around "Left Behind".
- `visual-figures-image-only`: Figure-formatting issues.
- `visual-pages`: Float / page-layout issue.
- `writing-intro`: Unneeded filler paragraphs, misplaced key material, tonal break caused by AI-authorship disclosure, main argument hard to follow on a skim.

These are minor, with the exception of `element-gkp-cites`. I totally would **not** put my name on a paper that mis-characterizes the key closely-related paper. `iter 5` fixed the problem.

The later iterations were focused on minor issues. For example:

- `iter 10`
  - Attempted but failed: Tighten intro flow.
  - Fixed: Figure 2 readability, Prop 2 (iii) stationarity discussion.
- `iter 11`
  - Attempted but failed: Fix extension figure whitespace.
  - Fixed: Seru is not a co-author on Kogan-Papanikolaou-Stoffman 2020, negative dividend accounting problem, `writing-intro` flow and contribution clarity.

The next 25 iterations were mostly failure churn. There would be 1-4 failures that would bounce around across tests, though `writing-intro` was a consistent blocker. The failures were rarely factchecks, so I feel one could have stopped the loop at any point after `iter 10` and just polished the rest by hand.

The final iterations were:

- `iter 35`
  - Remaining failure: `visual-figures-image-only`, because the extension figure still had visual-formatting issues.
- `iter 36`
  - Remaining failures: `spec-paper`, because Proposition 3's proof was still too long to remain inline; and `writing-intro`, because the complete-markets counterfactual was still not visible enough to a skimming reader.

At this point, I gave up. On my two $200/month Claude Code subscriptions, I could only run about 6-8 iterations per day, and it was unclear these iterations were going to pass all tests.

In fact, I felt I had to intervene in the "final" paper draft, as the abstract had this bizarre sentence "The displacement the paper models is closer than it appears." It also overclaimed in saying "All analysis, code and prose were produced by AI agents." This last, final edit is documented in this commit.

References

Acemoglu, Daron (2025). "The Simple Macroeconomics of AI". In: *Economic Policy* 40.121, pp. 13–58.

Aghion, Philippe, Benjamin F. Jones, and Charles I. Jones (2019). "Artificial Intelligence and Economic Growth". In: *The Economics of Artificial Intelligence: An Agenda*. University of Chicago Press, pp. 237–290.

Babina, Tania et al. (2024). *Artificial Intelligence and Firms' Systematic Risk*. Available at SSRN 4868770. DOI: 10.2139/ssrn.4868770.

Barro, Robert J. (2006). "Rare Disasters and Asset Markets in the Twentieth Century". In: *Quarterly Journal of Economics* 121.3, pp. 823–866.

Bengio, Yoshua (2023). "AI and Catastrophic Risk". In: *Journal of Democracy* 34.4, pp. 111–121.

Chen, Andrew Y., Alejandro Lopez-Lira, and Tom Zimmermann (2022). *Does Peer-Reviewed Research Help Predict Stock Returns?* arXiv: 2212.10317 [q-fin.PM].

Gârleanu, Nicolae, Leonid Kogan, and Stavros Panageas (2012). "Displacement Risk and Asset Returns". In: *Journal of Financial Economics* 105.3, pp. 491–510.

Hadfield-Menell, Dylan and Gillian K. Hadfield (2018). *Incomplete Contracting and AI Alignment*. arXiv: 1804.04268 [cs.AI].

Jones, Charles I. (2024). "The AI Dilemma: Growth versus Existential Risk". In: *American Economic Review: Insights* 6.4, pp. 575–590.

Knesl, Jiří (2023). "Automation and the Displacement of Labor by Capital: Asset Pricing Theory and Empirical Evidence". In: *Journal of Financial Economics* 147.2, pp. 271–296.

Kogan, Leonid and Dimitris Papanikolaou (2014). "Growth Opportunities, Technology Shocks, and Asset Prices". In: *Journal of Finance* 69.2, pp. 675–718.

Kogan, Leonid, Dimitris Papanikolaou, and Noah Stoffman (2020). "Left Behind: Creative Destruction, Inequality, and the Stock Market". In: *Journal of Political Economy* 128.3, pp. 855–906.

Korinek, Anton and Donghyun Suh (2024). *Scenarios for the Transition to AGI*. arXiv: 2403.12107 [econ.GN].

Lu, Chris et al. (2024). *The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery*. arXiv: 2408.06292 [cs.AI].

Nordhaus, William D. (2021). "Are We Approaching an Economic Singularity? Information Technology and the Future of Economic Growth". In: *American Economic Journal: Macroeconomics* 13.1, pp. 299–332.

Novy-Marx, Robert and Mihail Velikov (2026). "Artificial Intelligence-Powered (Finance) Scholarship". In: *Journal of Economic Literature* 64.1, pp. 5–37. DOI: 10.1257/jel.20251821.

Pástor, Ľuboš and Pietro Veronesi (2009). "Technological Revolutions and Stock Prices". In: *American Economic Review* 99.4, pp. 1451–1483.

Wachter, Jessica A. (2013). "Can Time-Varying Risk of Rare Disasters Explain Aggregate Stock Market Volatility?" In: *Journal of Finance* 68.3, pp. 987–1035.

Zhang, Miao Ben (2019). "Labor-Technology Substitution: Implications for Asset Pricing". In: *Journal of Finance* 74.4, pp. 1793–1839. DOI: 10.1111/jofi.12766.